



What determines liquidity?

SMM921: Trading and Microstructure

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$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

Roadmap

Setting: you are a trader or portfolio manager. You have to trade a particular set of stocks over the coming days. You want to estimate how much you will pay in trading costs.

You will want to know:

- How illiquidity and costly trading can be driven by **information asymmetries**
- How information asymmetries contribute to lack of depth and high bid-ask spreads — and how to measure their magnitude
- How **dealer risk aversion** affects trading costs, spreads, and depth
- How both channels combine to produce **severe illiquidity** in market stress episodes

Note on coverage: The full mathematical derivations of the models we present here are available in the **Appendix** for those who wish to work through them in detail. The main slides focus on mechanisms, results, and intuition.

Asymmetric Information and Securities Trading

Basic setup;

- Set of competitive dealers provide bid and ask quotes.
- Dealers are risk neutral
- Dealers face **information risk** or **adverse selection risk**;
 - This is the risk of a dealer trading with someone better informed than he is.
 - Some (not all) traders have private information that the dealer does not have.

market for lemons arkelof

if someone new gets into the
market the price will move up
bc its an information story
and not just a demand story

Implications;

- Trades cause prices to move **permanently**: prices rise (fall) after buys (sells).
- As the proportion of informed traders increases, markets become less liquid.
 - In some models, more informed traders leads to lower depth
 - In others, more asymmetric information leads to greater spreads

Asymmetric Information and liquidity: Glosten and Milgrom (1985)

Glosten and Milgrom (1985) have a **sequential** model of trading in an asset;

- Individuals arrive one at a time at a dealer's post and choose whether to buy or sell a unit of the asset at the posted bid and offer quotes (or not to trade at all).
- Dealer can't tell if an agent is informed or uninformed.
- All agents in the model are assumed to be risk-neutral.

Glosten and Milgrom (1985): Assumptions

Assumptions:

μ share of informed buyers
the non smart are flipping a coin and the smart traders will take the right one
"noise traders" or "liquidity traders"
and then either you repeat and get a stock movement

- **Single risky asset:** terminal payoff V is 0 or 1 with equal probability.
- **Liquidity traders:** population size is $1 - \mu$ and trade for unspecified reasons. Equally likely to buy and sell.
- **Informed traders:** population size is μ . They know the terminal asset payoff and trade so as to profit from this information.
- **Dealer:** risk neutral. Can't distinguish informed traders from liquidity traders. Faces threat of competition so expected profit is zero.
- **Timing:**
 - Trader is randomly chosen from entire population of traders and arrives at market.
 - Dealer quotes bid and ask prices (for single unit).
 - Trader decides whether and in which direction to trade.

true value is revealed after
both players play

Glosten and Milgrom (1985): the demand side of the market

Start with a a single period model where, after one trade, the true asset value is revealed.

How does each type of trader behave when he arrives at the dealing post?

Informed trader: can condition their decision on the **actual** asset payoff. Assuming that the bid and ask quotes set by the dealer are between 0 and 1 we get the following;

- If $V = 1$ they will buy with probability 1. knowledge so 1 shot
- If $V = 0$ they will sell with probability 1.

Liquidity trader: trades randomly and independently of the value of V . Thus;

- If $V = 1$ they will buy with probability $\frac{1}{2}$ and sell with probability $\frac{1}{2}$.
- If $V = 0$ they will buy with probability $\frac{1}{2}$ and sell with probability $\frac{1}{2}$.

coin flip

if played infinitely and the price will go to V since both half of liquidity traders and all informed traders will lead the price to V

Glosten and Milgrom (1985): the dealer's problem [1]

We know that;

- The dealer can't distinguish informed from liquidity traders.
 - So she can't condition quotes on the trader type.
- The dealer can quote a different price to buyers than she does to sellers, though.
- The dealer must only break even on average due to potential competition and risk neutrality.

Hence the dealer sets bid and ask quotes as follows;

$$P_b = E[V | \text{sell}] \quad , \quad P_a = E[V | \text{buy}]$$

These are **regret free** quotes in that they are prices which the dealer thinks are fair.

if someone reaches me and ask what is the asset is worth what would i value my asset at?
and the ask bid spread will be higher if the value will be higher or lower

Glosten and Milgrom (1985): the key mechanism [1]

The dealer cannot distinguish informed from uninformed traders. She must quote the **same bid and ask to everyone**.

What does the direction of a trade reveal?

is the new buyer bringing in more information?
if prices move a lot there is a lot of asymmetric info

the order flow to see how much price changes

- If $V = 1$: informed traders *always* buy; liquidity traders buy or sell randomly
 - If $V = 0$: informed traders *always* sell; liquidity traders buy or sell randomly
- ⇒ A **buy order** is more likely to come from an informed trader when $V = 1$ than when $V = 0$
- ⇒ The dealer updates her belief toward $V = 1$ upon seeing a buy, and **raises her ask**
- ⇒ Symmetrically, a sell order depresses the bid

Glosten and Milgrom (1985): the key mechanism [2]

the more informed traders the higher the spread

uninformed buyers are kicked out

no one wants to participate in a market if the other person has more info

The key result: with proportion μ of informed traders, determines the equilibrium bid-ask (in the simple model presented here **spread** = μ)

- More informed traders ($\mu \uparrow$) $\rightarrow$ wider spread $\rightarrow$ less liquidity
- If there are no informed traders ($\mu = 0$): spread is zero – no adverse selection, no reason to quote a spread
- If *all* traders are informed ($\mu = 1$): the market breaks down entirely

[**Note:** The full model derivation via Bayes' rule is in the appendix.]

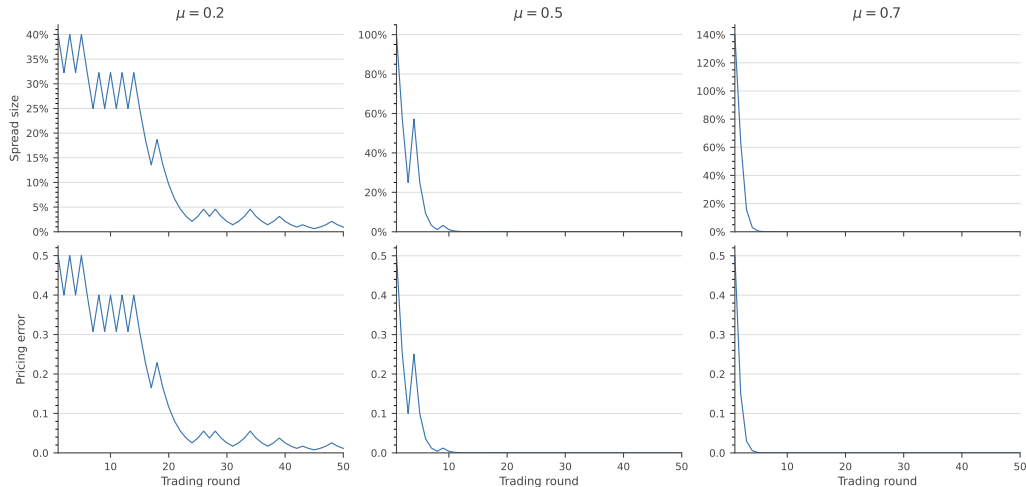
Glosten and Milgrom (1985): important results

- Bid and ask quotes **straddle** the dealer's best estimate of value – the midquote is the dealer's posterior mean
- The **bid-ask spread** arises for purely informational reasons. The spread will:
 - **Increase** with the proportion of informed traders (μ)
 - **Increase** with the precision of the informed traders' private information
- In multi-period models, the sequence of transaction prices follows a **martingale**:
$$E_t(P_{t+1} \mid P_t) = P_t$$
- Transactions cause quotes to move **permanently** (as they carry information)
- Prices converge to full-information levels after many trading rounds
- In some generalisations, asymmetric information can be so severe that the market **breaks down entirely**

Glosten-Milgrom (1985): many periods

price becomes too efficient and no one wants to participate
again price discovery

The x-axis counts trading rounds. The left-hand plots shows what happens if 20% of the population is informed, the middle is 50% informed and the right is 70% informed.



A complementary view: Kyle (1985) [1]

Glosten–Milgrom shows how asymmetric information **widens spreads**. Kyle (1985) shows how it **reduces depth** – and introduces a measure that practitioners use directly.

Key differences from Glosten–Milgrom:

- A **single** strategic informed trader, not a pool
- The informed trader trades **gradually**, disguising orders within noise trader flow to avoid revealing information too quickly
- Market maker prices based on **aggregate** signed order flow, not individual orders

A complementary view: Kyle (1985) [2]

The key result: Kyle's λ : the equilibrium pricing rule is:

$$\Delta p = \lambda \cdot x$$

diff btw buyers and sellers

where Δp is the change in the traded price, x is net signed order flow and λ measures **price impact per unit of order flow**.

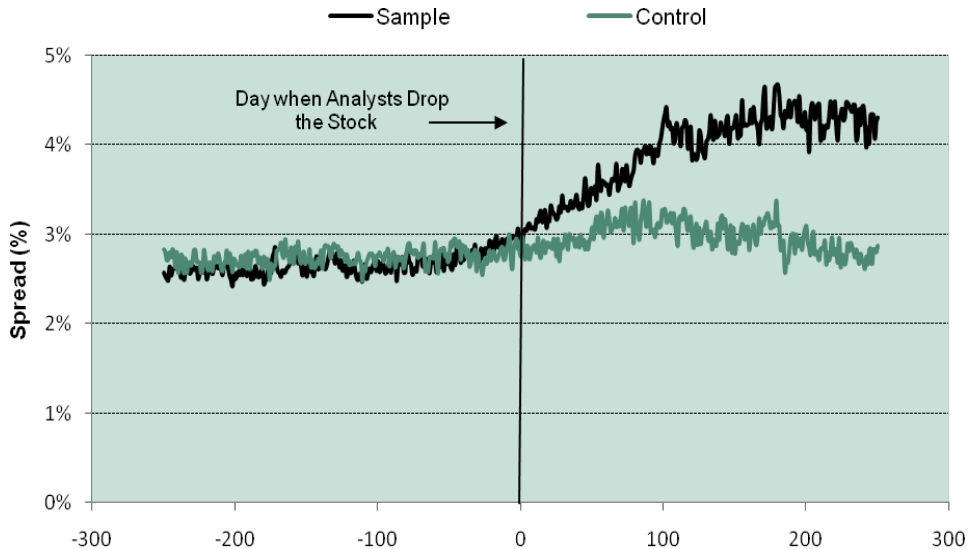
Comments:

not speed big lambda responsiveness of the market
losses are absorbed by a lot or not

- High $\lambda \rightarrow$ illiquid market: small trades move prices a lot
- Low $\lambda \rightarrow$ liquid market: large trades absorbed with little price impact
- λ **increases** with quality of private information and **decreases** with noise/liquidity trading intensity.
- Practitioners estimate λ directly as an input to execution algorithms and market impact models

you can predict the lambda
from getting 500 stocks and

Analyst coverage and liquidity: why do spreads widen when analyst coverage falls?



Asymmetric information: summary of key results

What have we learned from these asymmetric information models?

- **Prices respond to trades** — investor buys move prices up and sales reduce prices **permanently**, because order direction carries information about fundamentals
- **More asymmetric information, less liquidity:**
 - **Glosten–Milgrom:** more informed traders → **wider spreads** (spread = μ , the informed trader proportion)
 - **Kyle (1985):** more private information → **lower depth** (higher λ , greater price impact per unit traded)
 - Both models help explain why small-cap stocks are *less liquid* than large-caps: less analyst coverage → more information asymmetry → wider spreads and higher price impact
- **More noise traders** → **better liquidity:** uninformed flow provides “cover” for the dealer, reducing adverse selection costs and compressing spreads
- **For the practitioner:** the scale of the information asymmetry problem in a particular stock will directly affect your trading costs. Kyle’s λ , estimated from intraday data, gives you a **measurable handle** on this.

Risk averse dealers and pricing

Now change focus to concentrate on another aspect of a dealer's problem.

- Assume completely symmetric information.
- But dealers are now risk averse
 - This means that they don't want to be too long or too short stock.
- They set prices at which others can trade, but taking the other side of those trades means that they are taking unwanted risky positions
- Dealers thus charge prices that compensate them for dis-utility associated with having to bear **inventory risk**.
- This leads to illiquidity and positive **bid-ask spreads**

the bid ask spread for a risk adverse dealer is not good
and aversion leads to illiquidity
essentially im offering free options so as the world becomes more volatile the option value
increases and the dealer increases spreads

Inventory control: model [1]

Starting assumptions:

- Two period problem: trade today (t) and uncertainty about value of risky asset resolved tomorrow ($t + 1$)
- Dealer has mean-variance preferences defined over $t + 1$ wealth:

$$U(w_{t+1}) = E_t(w_{t+1}) - \frac{\rho}{2} \text{Var}_t(w_{t+1})$$

where w_{t+1} is wealth

- ρ measures dealer risk aversion.

utility is in increasing in wealth and decreasing in variance
so adverse and concave

Inventory control: model [2]

- Dealer has cash worth c_t at t and has z_t units of stock at t
- Thus dealer wealth at t is

budget constraint
stock at a price + cash

$$w_t = p_t z_t + c_t$$

where p_t is the current price of stock.

- Stock liquidates tomorrow with value (v_{t+1}) drawn from a normal distribution;

$$v_{t+1} \sim N(\mu, \sigma^2)$$

normal assumption

Inventory control: the dealer's supply decision [1]

The dealer chooses how much stock y_t to sell to the market at the prevailing price p_t , subject to her mean-variance preferences.

Maximising utility over next-period wealth gives the **optimal supply function**:

$$y_t = \frac{p_t - \mu}{\rho\sigma^2} + z_t$$

optimal amount to sell =
(price today - best guess
tomorrow)/(risk aversion)
+ amount of stocks

Economic interpretation:

- If $p_t = \mu$: dealer sells *entire* inventory — holding stock earns no premium, but entails bearing risk
in the absence of speculative cash is better
+ z sale
- If $p_t > \mu$: dealer supplies *more* than inventory, taking a short position — the premium compensates for risk
more than fair value so sell everything and take a short position
- If $p_t < \mu$: dealer holds back, maintaining a long position keep some
- Higher risk aversion ($\rho \uparrow$) or higher volatility ($\sigma^2 \uparrow$): dealer is less willing to hold any inventory, so the quantity supplied is *less sensitive* to the price — less liquidity provided

Inventory models: equilibrium [1]

as a dealer i cant just sell or buy so i need to set my prices to match my y to the price
makert equilibriium

Equilibrium: we must have demand equals supply. If demand for the stock is D_t (where D_t is above zero for a customer buy and below zero for a sell) then $D_t = y_t$ implies that the price is;

$$p_t = \mu - \rho\sigma^2 z_t + \rho\sigma^2 D_t$$

$D > 0$ buying
 $D < 0$ selling
more demand is sitll higher price
even with risk aversion i still get the same as
demand and supply at the beginning
with risk aversion from the dealer

The first two terms are independent of the incoming trade so define $m_t = \mu - \rho\sigma^2 z_t$. Then we have;

$$p_t = m_t + \rho\sigma^2 D_t$$

Inventory models: equilibrium [2]

illiquidity is a function of risk
adversion and volatility of the
world

Liquidity: measure market liquidity by the bid-ask spread. This is the price the dealer charges for selling one share ($D_t = +1$) less the price he charges for buying one share ($D_t = -1$). Our pricing equation tells us that this is;

$$S_t = 2\rho\sigma^2 \quad \text{spread}$$

Thus;

- Spreads increase (liquidity decreases) with dealer risk aversion
- Spreads increase (liquidity decreases) with the volatility of the stock

As you can also see from the previous pricing equation, as trade sizes get larger prices move further (i.e. the model determines depth too).

Inventory control: summary

- Risk aversion leads to another channel for liquidity determination.
- Liquidity drops (spreads rise) when markets become more volatile
- Cost of trading also naturally larger for larger trades (i.e. risk aversion leads to market impact)
- It's not in the simple model, but spreads also rise as trading frequency drops.
- **Trades move prices** again.
 - Buys from the dealer move prices up and sells to the dealer move prices down
 - The same directional relationship as in asymmetric information models.
- In this setting, though, price moves in response to trades are **temporary**
 - They disappear when the inventory imbalance has been eliminated.

Liquidity in market stress periods: when theory meets reality

The two channels we have studied – **information asymmetry** and **inventory risk** – typically operate quietly in well-functioning markets.

if there is a moment of volatility all at once there is some gains in speed for the market incorporating new info

In a **market crisis**, both channels amplify simultaneously and may reinforce each other:

- **Information channel:** extreme uncertainty about fundamentals → adverse selection risk for market makers rises → spreads widen sharply as dealers price in the risk of trading with better-informed counterparties
- **Inventory channel:** volatility increases ($\sigma^2 \uparrow$ in our model) → inventory risk becomes acute → dealers dramatically reduce willingness to hold positions → depth collapses
- **The interaction:** falling prices force leveraged investors to sell, increasing order imbalance, which tests already stretched dealer capacity – a **self-reinforcing cycle**

inventory capacity also means more liquidity taken out of the market
both retail and sophisticated investors

This dynamic is why markets can seize up **rapidly**, even for assets that were highly liquid a day before.

Liquidity in market stress: March 2020 [1]

March 2020 can be seen as a very clear recent illustration of both channels operating simultaneously across multiple asset classes.

US equity markets:

- Bid-ask spreads on S&P 500 stocks widened to levels not seen since the 2008 financial crisis
- The VIX peaked at 66 on 16 March 2020 – the second-highest reading in its history
- HFT market makers, normally the dominant source of lit market liquidity, sharply reduced quoting activity as inventory risk became unmanageable (σ^2 exploded)

Liquidity in market stress: March 2020 [2]

US Treasury markets: the more striking episode

- Normally the world's most liquid securities
- Bid-ask spreads on on-the-run Treasuries widened by **10x** within days
- Primary dealers – banks obligated to make markets – hit balance sheet limits and were unable to absorb selling pressure
- The Federal Reserve ultimately purchased \$1.6 trillion of Treasuries to restore market function

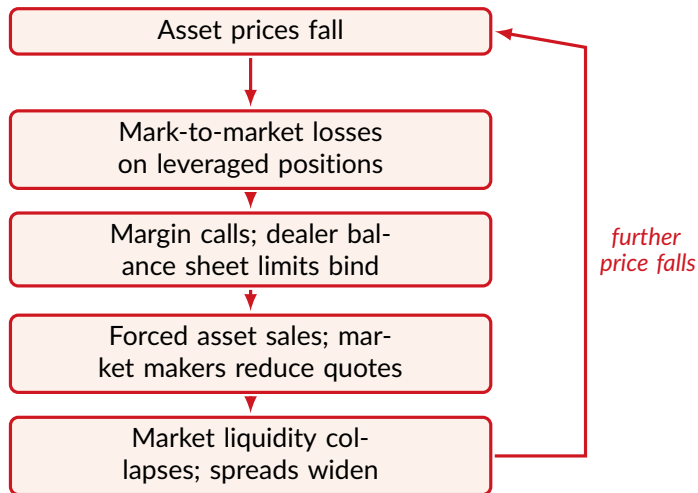
In sum: both channels were operating:

- **information** (extreme uncertainty about pandemic trajectory)
- **and inventory** (dealer balance sheets overwhelmed, volatility at historic highs)

all tourism what is it worth
now?

Liquidity spirals [1]

Brunnermeier/Pedersen (2009): market and funding liquidity create self-reinforcing cycles.



Liquidity spirals [2]

Policy implication: central bank interventions in March 2020 (Federal Reserve) and March 2023 (Fed/FDIC in the SVB episode) were specifically designed to **break the spiral** by providing funding liquidity to dealers and reducing forced-selling pressure.

Overall Summary [1]

What should you worry about when trading? What determines market liquidity? What affects your trading costs?

- **Information risk:**

- Trading costs are greater in markets with many informed traders
- Your trades move prices **permanently** — other market participants infer information from your order
- Kyle's λ gives a practical, measurable handle on price impact driven by information asymmetry

- **Price/inventory risk:**

- Trading costs are greater when volatility is high ($\sigma^2 \uparrow$)
- Trading costs are greater with infrequent trading and high effective dealer risk aversion ($\rho \uparrow$)
- Larger trades cost more: market impact grows with order size
- Price moves from inventory are **temporary**: they reverse once the inventory imbalance is eliminated

Overall Summary [2]

- **In a crisis, both channels amplify simultaneously:**
 - Uncertainty surges, dealer balance sheets bind, forced selling cascades
 - **Liquidity spirals** can cause rapid, severe disruption even in normally liquid assets
 - March 2020 in US equities and Treasury markets is the defining recent example

Appendix

Appendix: Glosten–Milgrom – the dealer's problem [1]

To compute optimal dealer quotes we make use of **Bayes' rule** and the following equations;

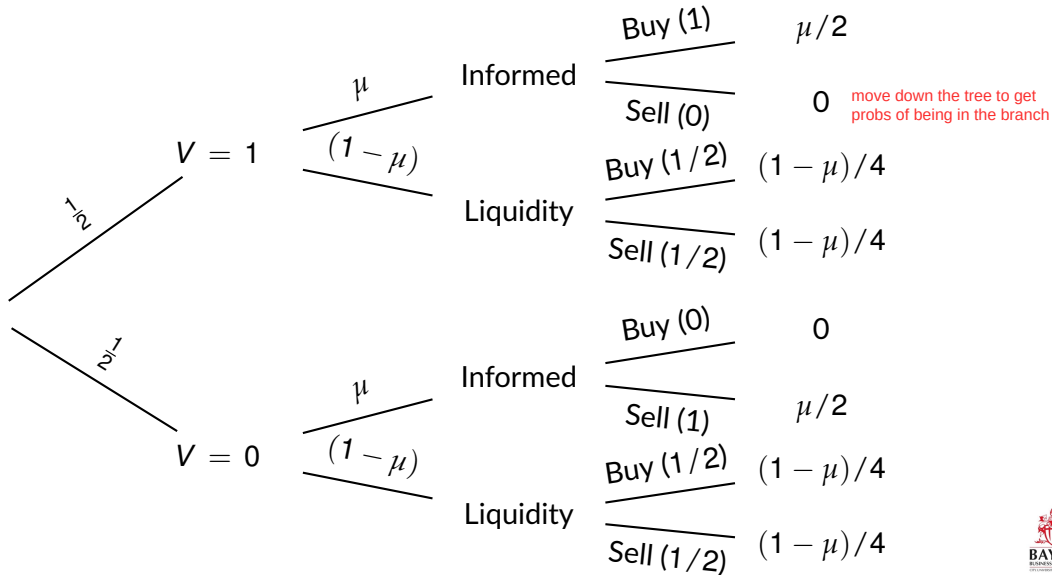
$$\begin{aligned} E[V | \text{sell}] &= 1 \cdot \Pr[V = 1 | \text{sell}] + 0 \cdot \Pr[V = 0 | \text{sell}] = \Pr[V = 1 | \text{sell}] \\ E[V | \text{buy}] &= 1 \cdot \Pr[V = 1 | \text{buy}] + 0 \cdot \Pr[V = 0 | \text{buy}] = \Pr[V = 1 | \text{buy}] \end{aligned}$$

if the true value is 0 no one will buy and everyone would sell

if the true value is 1 no one will sell and everyone would buy

A summary of the informational and probabilistic structure of the model is given in the following diagram. Nature/God chooses the realisation of V and then randomly picks a trader. The trader then submits a market order.

Appendix: Glosten-Milgrom – the dealer's problem [2]



Appendix: Glosten-Milgrom: optimal bid/offer

Bayes' rule tells us that the probability of B conditional on A is the joint probability of B and A over the total probability of A . Using the prior diagram then implies;

$$P_a = \Pr[V = 1 \mid \text{buy}] = \frac{\Pr[V = 1 \cap \text{buy}]}{\Pr[\text{buy}]} = \frac{\frac{\mu}{2} + \frac{(1-\mu)}{4}}{\frac{\mu}{2} + \frac{(1-\mu)}{4} + \frac{(1-\mu)}{4}} = \frac{1}{2} + \frac{\mu}{2}$$
$$P_b = \Pr[V = 1 \mid \text{sell}] = \frac{\Pr[V = 1 \cap \text{sell}]}{\Pr[\text{sell}]} = \frac{\frac{(1-\mu)}{4}}{\frac{(1-\mu)}{4} + \frac{\mu}{2} + \frac{(1-\mu)}{4}} = \frac{1}{2} - \frac{\mu}{2}$$

Obviously: as long as informed traders exist, the offer price will be larger than the bid price i.e. there will be a **positive bid-ask spread** and it is equal to μ .

Appendix: Inventory control — derivation [1]

Assume: dealer takes price (p_t) of stock as given.

- Define y_t to be the amount of stock that the dealer will supply to the public for sale at price p_t
- If he sells y_t shares, his $t + 1$ inventory becomes;

$$Z_{t+1} = Z_t - y_t$$

and his cash stock is;

$$C_{t+1} = C_t + p_t y_t$$

- So we can write wealth at time $t + 1$ as;

$$W_{t+1} = v_{t+1}(Z_t - y_t) + C_t + p_t y_t$$

Appendix: Inventory control – derivation [2]

Maximise utility: the dealer chooses y_t to maximise;

$$\begin{aligned} U(w_{t+1}) &= E_t(w_{t+1}) - \frac{\rho}{2} \text{Var}_t(w_{t+1}) \\ &= \mu(z_t - y_t) + c_t + p_t y_t - \frac{\rho}{2} \sigma^2 (z_t - y_t)^2 \end{aligned}$$

The first-order condition is;

$$p_t - \mu + \rho \sigma^2 (z_t - y_t) = 0$$

so;

$$y_t = \frac{p_t - \mu}{\rho \sigma^2} + z_t$$